From the table, we conclude that

- **A natural number greater than 1, having only two factors namely 1 and the number itself, is called a *prime number.***
  For example, 2 (1 x 2) is a prime number as is 13 (1 x 13).

Note

The number '1' is neither prime nor composite

- **A natural number having more than 2 factors is called a *composite number.***

  For example, 15 is a composite number (15 = 1 x 3 x 5) as is 70 (1 x 2 x 5 x 7).

DO YOU KNOW?

A number is a ***perfect number*** if the sum of its factors other than the given number gives the same number. For example, ***6 is a perfect number***, since adding the factors of 6 (other than 6), namely 1, 2 and 3 gives the given number 6. i.e.,1+2+3 = 6 is the given number.

Check whether 28, 54 and 496 are perfect numbers or not.

Activity

(i) List out the prime and composite numbers represented by the dates in the month of October.

(ii) Generate a few composite numbers by product of two or more natural numbers.

(iii) Classify the numbers 34, 57, 71, 93, 101, 111 and 291 as prime or composite.

October 2018

1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

1.2.1 Finding the Prime Numbers by Sieve of Eratosthenes Method

Sieve of Eratosthenes, is a simple method of elimination by which we can easily find the prime numbers upto a given number. This method given by a Greek Mathematician, Eratosthenes of Alexandria, follows some simple steps which are listed below, by which we can find the prime numbers.

Step 1: Create 10 rows and 10 columns and write the numbers from 1 to 10 in the first row, 11 to 20 in the second row and continue the same as 91 to 100 in the tenth row.

Step 2: Leave **1** as it is neither prime nor composite (Why?). Start with the smallest prime 2. Encircle and colour 2 and cross out all other multiples of 2 (all even numbers) in the grid.

Step 3: Now, take the next prime 3. Encircle and colour 3 and cross out all other multiples of 3 in the grid.

Step 4: As 4 is crossed out already, go for the next prime 5 and cross out multiples of 5, except 5.

Step 5: Keep doing this, for two more primes 7 and 11 and stop. (Think why?)

The above steps are carried out to find prime numbers upto 100 in the following grid.

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